

Isaac Manring

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Education

INDIANA UNIVERSITY PURDUE UNIVERSITY IN INDIANAPOLIS (IUPUI)

Bachelor of Science – Double major in Applied Statistics and Computer Science

8/19-5/22

- Member of the IUPUI Honors College
- GPA 3.99/4.0

Master's in Computational Data Science

8/22- 12/23

- GPA 4.0/4.0
- Masters project: Manring, I., Wang, H., Mohler, G., & Miscouridou, X. (2025). BSTPP: a python package for Bayesian spatiotemporal point processes. *Journal of Applied Statistics*, 1–20.
<https://doi.org/10.1080/02664763.2025.2462969>

UNIVERSITY OF FLORIDA

PhD in Computer Science

8/25-present

Honors/Awards

- Top 100 in IEEEEXTREME worldwide (2,992 participants) and 2nd in US
- Chancellor's Scholarship, Dean of Science Scholarship recipient, and Anna K Suter Scholarship.
- Dean's List fall 2019, spring 2020. Science Scholar's list fall 2020 to spring 2022
- IUPUI French Department, 2019-2020 Top Scholar in French award
- Outstanding Senior in Applied Statistics award.

Work Experience

ALLSTATE INSURANCE COMPANY

1/24-8/25, Remote

Data Scientist on the Telematics team

- Worked with telematics research team to build loss models and derive insights from users' trip data
- Extensive experience with distributed computing using PySpark, and H2o

ALLSTATE INSURANCE COMPANY

5/22-8/2022 and 5/23-8/23, Remote

Data Science Intern

- Developed a data pipeline with PySpark.
- Performed preliminary analysis for loss-cost modeling and interpreted current models in R.
- Built new geospatial features with SQL and GeoPandas for modeling.
- Devised and implemented novel predictive models to evaluate geospatial underwriting rules.

IUPUI DEPARTMENT OF COMPUTER SCIENCE

10/21-12/23, Indianapolis, IN

Research Assistant to Dr. George Mohler

- Built deep transformer models for audio event detection with Pytorch.
- Developed Bayesian sound source localization methodology.
- Implemented firmware for model deployment with quantization on Raspberry Pi.
- Manring, I., Hill, J., Mohler, G., Brantingham, P., Williams, T., White, B. (2023, October 9-13). Low-Cost Gunshot Detection System with Localization for Community Based Violence Interruption. DSAA 2023, Thessaloniki, Greece.

IUPUI DEPARTMENT OF MATHEMATICS

6/21-5/22, Indianapolis IN

Undergraduate Research Assistant to Dr. Luoding Zhu

- Coded fluid dynamics numerical methods for osteocyte model. Visualized results.

- Manring, I., Zhu, L., & Barber, J. (2022). 7th International Conference on Computational and Mathematical Biomedical Engineering. In SIMULATION OF FLOW-OSTEOCYTE INTERACTION IN A LACUNO-CANALICULAR NETWORK (1st ed., Vol. 2, pp. 683–686). Milan; Computational and scientific consultancy services Ltd.
- Barber, J., Manring, I., Boileau, S., & Zhu, L. (2023). Modeling and simulation of flow–osteocyte interaction in a lacuno-canalicular network. *Physics of Fluids*, 35(9). <https://doi.org/10.1063/5.0165467>

IUPUI DEPARTMENT OF COMPUTER SCIENCE

1/20-12/21, Indianapolis, IN

Teaching Assistant for Computing I, Computing II, Principles of Software Design

- Led computer science recitation and held office hours.

IUPUI MATH ASSISTANCE CENTER

1/20-10/21, Indianapolis, IN

Math Tutor

- Assisted students with homework for statistics, multivariable calculus, and differential equations.

Technical Skills

RELEVANT COURSEWORK: Deep Learning, Statistical Machine Learning, Natural Language Processing, Database Systems, Bayesian Statistics, Nonparametric Statistics, Regression Analysis, Point Processes.

TECHNOLOGIES AND PROGRAMMING LANGUAGES: Python, Pytorch, PySpark, Pandas, GeoPandas, R, SQL, Big Query, C++, Django, HTML

LINKS: <https://github.com/imanring>